Sungwon Kim

3900N Charles St. Baltimore, MD 21218 | 01) 917-454-8696 | skim434@jhu.edu

EDUCATION

Johns Hopkins University *Baltimore* ***Baltimore, MD***
- GPA: 3.89/4.0 (4.0/4.0 in all 400+ level courses) *Expected:December 2024*
- BS Applied Mathematics & Statistics, BS/ MsE Computer Science

Work Experience

Johns Hopkins University ***Baltimore, MD***
Course Assistant *January 2024 – Current*
- Course Assistant for NLP Self-Supervised Models

Johns Hopkins University ***Baltimore, MD***
Course Assistant *August 2023 – December 2023*
- Course Assistant for Natural Language Processing
- Graded, and helped with homework revisions.
- Held weekly office hours.

Center for Language and Speech Processing ***Baltimore, MD***
Research Assistant *May 2023 – Current*
- Created chatbots which refers to a database for responses.
- Use Combination of Dense / Sparse, and traditional retrievals for best results.
- Obtained metrics for different RAG approaches

Johns Hopkins University ***Baltimore, MD***
Research Assistant *May 2023 – September 2023*
- Created neural network-derived brain perfusion map processing for supplementary data on ischemic strokes.
- Achieved performance better than recent paper.

Johns Hopkins University ***Baltimore, MD***
Course Assistant *January 2023 – May 2023*
- Course Assistant for Introduction to Algorithms

Republic of Korea Army ***Yanggu, Korea***
Information Systems Maintenance/Operator, Squad Leader *September 2020 – March 2022*
- In charge of receiving and sending sensitive information
- Squad Leader from June of 2021

Nexon Korea ***Pangyo, Korea***
Platform Analyst *June 2020 – August 2020*
- Analyzed game data logs using SQL, Hadoop, and Python
- Distinguished experienced players from the playerbase using machine learning techniques including various clustering methods, random forest, LSTMs, and CNN

Johns Hopkins University ***Baltimore, MD***
Teaching Assistant/Course Assistant *January 2020 – May 2020*
- Teaching Assistant for Discrete Mathematics
- Course Assistant for Automata & Computation Theory

PROJECTS

arXiv Reference finder (Hophacks) *September 2023*
- Awarded second overall price for HopHacks 2023
- Given a research paper, finds relevant parts from the referenced papers, if the paper is in arXiv.

Custom Transformers *May 2023 – June 2023*
- Created customized transformer layers with non-standard implementations.
- Achieved better loss compared to Huggingface implementation of BERT.

Misc Game Tools(Toys) *June 2020 – Current*
- Created an excel sheet for calculating expected cost for equipment upgrades in computer games using markov chain
- Created a small planner which tracks every Johns Hopkins University course information locally.
- Coded Pokémon battle damage calculator & Derived method for type ranking calculation
- Coded Enigma machine and an encryption decoder using Index of Coincidence.

Othello Game *June 2019 – November 2019*
- Developed an Othello game and UI using python and wxpython.

- Implemented AI using minimax with alpha-beta pruning.

Randomized Game Tree *June 2017 – November 2017*

- Researched equations and algorithms that estimates the win rates of games given the random distribution of outcomes.
- Coded scenarios to verify results.
- Received a Siemens Competition Semifinalist Award.

SKILLS

Software/Skill set: LaTeX, SQL, Microsoft Office (Excel, PowerPoint, Word, Outlook)
Programming: Python, Pytorch, Huggingface Java, JavaScript, C, C++
Language: English (fluent), Korean (native)